MATH2012: Fundamental Concepts of Mathematics

Revision Class 1

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1. Key Concepts (Ch. 1 – 3.3)

2. Demonstration Examples

Definition – Elements and Subsets

(a) We write $a \in S$ to say that a is an element of set S.

(b) We write $A \subseteq B$ to say that every element in A is in B, which means that A is a subset of B.

- We can write (b) in a more mathematical way:

$$A \subseteq B \iff \forall x \in A, x \in B$$

- (Equality of Sets) $A = B$ **if and only if** $A \subseteq B$ and $B \subseteq A$.
- (Proper Subset) $A \subset B$ means $A \subseteq B$ and $A \neq B$.

Here, we let A and B to be subsets of a **universal** set U.

Definition – Set Operations

- Complement:
$$A^c := \{x \in U : x \notin A\}$$
- Union:
$$A \cup B := \{x : x \in A \text{ or } x \in B\}$$
- Intersection:
$$A \cap B := \{x : x \in A \text{ and } x \in B\}$$
- Symmetric difference:
$$A \triangle B = (A \setminus B) \cup (B \setminus A)$$

- Try: $A := \{x \in \mathbb{N} : x \text{ is divisible by } 2\}, B := \{x \in \mathbb{N} : x \text{ is prime}\}$

Example – $U = \mathbb{N}$

$A := \{x \in \mathbb{N} : x \text{ is divisible by } 2\}, B := \{x \in \mathbb{N} : x \text{ is prime}\}$

- Complement:

$$A^c = \{x \in \mathbb{N} : x \text{ is not divisible by } 2\}$$
$$B^c = \{x \in \mathbb{N} : x \text{ is composite or equal to } 1\}$$

- Union:

$$\begin{aligned} A \cup B &= \{x \in \mathbb{N} : x \text{ is prime or divisible by } 2\} \\ &= \{2, 3, 4, 5, 6, 7, 8, 10, 11, \ldots\} \end{aligned}$$

Example – $U = \mathbb{N}$

$A := \{x \in \mathbb{N} : x \text{ is divisible by } 2\}, B := \{x \in \mathbb{N} : x \text{ is prime}\}$

- Intersection:

$$\begin{aligned} A \cap B &= \{x \in \mathbb{N} : x \text{ is prime and divisible by } 2\} \\ &= \{2\} \end{aligned}$$

- Symmetric difference:

$$\begin{aligned} A \triangle B &= (A \setminus B) \cup (B \setminus A) \\ &= \{4, 6, 8, 10, \ldots\} \cup \{3, 5, 7, 11, \ldots\} \\ &= \{3, 4, 5, 6, 7, 8, 10, 11, \ldots\} \\ &\stackrel{?}{=} (A \cup B) \setminus (A \cap B) \end{aligned}$$

Definition – Power Sets and Cartesian Product

- Power Set:

$$\mathcal{P}(A) := \{S : S \subseteq A\}$$

- Cartesian Product:

$$A \times B := \{(a, b) : a \in A \text{ and } b \in B\}$$

- Let A be a finite set with $|A| = n$. Then $|\mathcal{P}(A)| = 2^n$.
- In general, $A \times B \neq B \times A$ unless $A = \emptyset$ or $B = \emptyset$ or $A = B$.

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- Let A be a finite set with $|A| = n$. Then $|\mathcal{P}(A)| = 2^n$.
- In general, $A \times B \neq B \times A$ unless $A = \emptyset$ or $B = \emptyset$ or $A = B$.
- In other MATH courses, you may see notations like

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} := \{(x, y) : x \in \mathbb{R} \text{ and } y \in \mathbb{R}\}$$

and hence see notations like $(x, y) \in \mathbb{R}^2$.

Theorem – Distributive Laws

- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Theorem – De Morgan's Laws

- $U \setminus (A \cup B) = (U \setminus A) \cap (U \setminus B)$
  - also written as $(A \cup B)^c = A^c \cap B^c$
- $U \setminus (A \cap B) = (U \setminus A) \cup (U \setminus B)$
  - also written as $(A \cap B)^c = A^c \cup B^c$

Theorem – Principle of Inclusion-Exclusion

- $|A \cup B| = |A| + |B| - |A \cap B|$
- $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$

- (Memorization Trick) We include all possible "intersection" of 1 sets, exclude all possible intersection of 2 sets, include all possible intersection of 3 sets, and so on.
- Try: Find a formula for $|A \cup B \cup C \cup D|$.
- Using the above trick:

$$
\begin{aligned}
&|A \cup B \cup C \cup D| = |A| + |B| + |C| + |D| \\
&\underbrace{-|A \cap B| - |A \cap C| - |A \cap D| - |B \cap C| - |B \cap D| - |C \cap D|}_{\text{all possible intersection of 2 sets}} \\
&+ |A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| + |B \cap C \cap D| \\
&- |A \cap B \cap C \cap D|
\end{aligned}
$$

Example – Principle of Inclusion-Exclusion

Take $X = \{1, 2, 3, \ldots, 1000\}$ and
$S = \{x \in X : x \text{ is a multiple of 3 or 5 or 12}\}$. What is the cardinality of S?

- Denote

$$A = \{x \in X : x \text{ is a multiple of } 3\}$$
$$B = \{x \in X : x \text{ is a multiple of } 5\}$$
$$C = \{x \in X : x \text{ is a multiple of } 12\}$$

- We have $|S| = |A \cup B \cup C| = |A \cup B|$ as $C \subseteq A$.
- As
$|A| = \left\lfloor \frac{1000}{3} \right\rfloor = 333, |B| = \left\lfloor \frac{1000}{5} \right\rfloor = 200, |A \cap B| = \left\lfloor \frac{1000}{15} \right\rfloor = 66$:

$$|S| = 333 + 200 - 66 = 467$$

Ch. 2 – Logic

A **statement** is a sentence with one and only one truth value, True (T) or False (F). We can construct **truth tables** for statements by listing out all possible combinations of 'true's and 'false's.

Definition (Combining statements)

- (Conjunction) $P \wedge Q$ is true if both P and Q are true; otherwise false.
- (Disjunction) $P \vee Q$ is false if both P and Q are false; otherwise true.
- (Negation) $\sim P$ has the opposite truth value of P.

Definition (Conditionals)

The statement "If P then Q" is denoted by $P \to Q$. P is called the hypothesis, and Q is called the consequent of the conditional.

- Notice that $P \to Q$ is false only when P is true and Q is false.
- If the hypothesis is false, the conditional is automatically true.

Definition (Converse and Biconditional)

- $Q \to P$ is called the converse of $P \to Q$.
- The biconditional $P \leftrightarrow Q$ means $(P \to Q) \wedge (Q \to P)$.

Definition (Implication)

$P \Rightarrow Q$ means the statement $P \to Q$ is true. Equivalently,

- P is **sufficient** for Q.
- Q is **necessary** for P.

Definition (Logical Equivalence)

$P \Leftrightarrow Q$ means both $P \Rightarrow Q$ and $Q \Rightarrow P$. Equivalently, we have

- P if and only if Q.
- P is necessary and sufficient for Q.

Theorem – Distributive Laws

- $P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$
- $P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$

- Compare with the set version:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Theorem – De Morgan's Laws

- $\sim (P \vee Q) \Leftrightarrow (\sim P) \wedge (\sim Q)$
- $\sim (P \wedge Q) \Leftrightarrow (\sim P) \vee (\sim Q)$

- Compare with the set version:

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

Notable Logical Equivalences

- (Implication) $P \longrightarrow Q \Longleftrightarrow (\sim P) \vee Q$
- (Contrapositive) $(\sim Q) \longrightarrow (\sim P) \Longleftrightarrow P \longrightarrow Q$

- The logical equivalences shown are very useful in some intermediate steps of proofs!

Now, we review a few concepts and notations about quantifiers and negations, which will be widely used in the later part of this course (and also other math courses).

- universal quantifier $\forall$
- existential quantifier $\exists$
- $\sim (\forall x \in S, P(x)) \Leftrightarrow \exists x \in S, \sim P(x)$
- $\sim (\exists x \in S, P(x)) \Leftrightarrow \forall x \in S, \sim P(x)$
- Note that the order of the quantifiers is important!
  $\forall x \in \mathbb{Z},\ \exists y \in \mathbb{Z},\ y > x$
  $\exists y \in \mathbb{Z},\ \forall x \in \mathbb{Z},\ y > x$

Example – Contrapositive

Find the contrapositive of the following statement:
P : If f is differentiable, then f is continuous.

- $\sim P$: If f is not continuous, f is not differentiable.
- Is this true?

Example – Negation

Find the negation of the following statement:
P : For any right $\triangle ABC$ with integer side lengths a, b, c, we have $a^2 + b^2 = c^2$ and an integer area.

- $\sim P$: There exists a right $\triangle ABC$ with integer side lengths a, b, c such that $a^2 + b^2 \neq c^2$ or a non-integer area for $\triangle ABC$.
- Is this true?

Ch. 3 – Techniques of Proof

For the coverage of Test 1, you are expected to know the following ways to prove mathematical statements:

1. Direct Proof
   - Involving Congruence of Integers (e.g. divisibility properties, modular arithmetic)
   - Sets (e.g. distributive property, De Morgan's Law)
2. Proof by Contrapositive
   - Resulting from $P \longrightarrow Q \Longleftrightarrow (\sim Q) \longrightarrow (\sim P)$
   - Usually used when Q is "easier" than P.
3. Proof by Cases
   - odd/even parity (more generally, going thru all possible $r \pmod{n}$)
   - different cases for being in parts of different subsets/sets

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2. Demonstration Examples

Demonstration Examples (Pre-class Q1)

Question – Write the set corresponding to the following:

(i) $\{\{a\}, \{a, b, c\}, a, b, c\} \setminus \{a\}$

(ii) $\{\{a\}, \{a, b, c\}, a, b, c\} \setminus \{a, b\}$

(iii) $\{\{a\}, \{a, b, c\}, a, b, c\} \setminus \{\{a\}, \{b, c\}\}$

(i) $\{\{a\}, \{a, b, c\}, b, c\}$

(ii) $\{\{a\}, \{a, b, c\}, c\}$

(iii) $\{\{a, b, c\}, a, b, c\}$

Demonstration Examples (Pre-class Q2)

Question – Find nonempty sets X, Y, A, B, C, D such that:

(i) $\mathcal{P}(X \setminus Y) \neq (\mathcal{P}(X) \setminus \mathcal{P}(Y)) \cup \{\emptyset\}$

(ii) $(A \times B) \cup (C \times D) \neq (A \cup C) \times (B \cup D)$

(i) We try to think what type of sets are bad.

- If $X \cap Y$ is empty, then $X \setminus Y = X$, $\mathcal{P}(X) \setminus \mathcal{P}(Y) = \mathcal{P}(X) \setminus \{\emptyset\}$ (not good for us!).
- Let $X = \{1, 2\}, Y = \{2\}$:

$$\begin{aligned} \mathcal{P}(X \setminus Y) &= \mathcal{P}(\{1\}) \\ &= \{\emptyset, \{1\}\} \end{aligned}$$

$$\begin{aligned} (\mathcal{P}(X) \setminus \mathcal{P}(Y)) \cup \{\emptyset\} &= \{\emptyset, \{1\}, \{2\}, \{1, 2\}\} \setminus \{\emptyset, \{2\}\} \cup \{\emptyset\} \\ &= \{\emptyset, \{1\}, \{1, 2\}\} \end{aligned}$$

Demonstration Examples (Pre-class Q2)

Question – Find nonempty sets X, Y, A, B, C, D such that:

(i) $\mathcal{P}(X \setminus Y) \neq (\mathcal{P}(X) \setminus \mathcal{P}(Y)) \cup \{\emptyset\}$

(ii) $(A \times B) \cup (C \times D) \neq (A \cup C) \times (B \cup D)$

(ii) We try to think what type of sets are bad.

- If $(x, y) \in (A \times B) \cup (C \times D)$, then $(x, y) \in A \times B$ or $(x, y) \in C \times D$.
- We potentially miss out on cases where $x \in C$ and $y \in B$!
- Consider $A = \{1\}, B = \{2\}, C = \{3\}, D = \{4\}$:

$$\begin{aligned}
(A \times B) \cup (C \times D) &= \{(1,2)\} \cup \{(3,4)\} \\
&= \{(1,2),(3,4)\} \\
(A \cup C) \times (B \cup D) &= \{1,3\} \times \{2,4\} \\
&= \{(1,2),(1,4),(3,2),(3,4)\}
\end{aligned}$$

Demonstration Examples (Pre-class Q3)

Question

Let P, Q and R be statements. Is

$$(P \longrightarrow Q) \text{ logically equivalent to } (R \wedge P) \longrightarrow (R \wedge Q)?$$

Show your work.

- <u>Method 1</u>: We try to go through all possible truth values of P, Q, R and construct the truth table (we know there are $2^3 = 8$ possible combinations).
- <u>Method 2</u>: We try to use some properties to transform $(R \wedge P) \longrightarrow (R \wedge Q)$ to potentially see parts of $(P \longrightarrow Q)$ $(\Leftrightarrow (\sim P) \vee Q)$.

Demonstration Examples (Pre-class Q3)

- Method 1: We can construct a truth table:

P	Q	R	$P \to Q$	$R \wedge P$	$R \wedge Q$	$(R \wedge P) \to (R \wedge Q)$
T	T	T	T	T	T	T
T	T	F	T	F	F	T
T	F	T	F	T	F	F
T	F	F	F	F	F	T
F	T	T	T	F	T	T
F	T	F	T	F	F	T
F	F	T	T	F	F	T
F	F	F	T	F	F	T

From the truth values in the 4th row, we see that the two statements are NOT logically equivalent.

Demonstration Examples (Pre-class Q3)

- <u>Method 2</u>: We transform $(R \wedge P) \longrightarrow (R \wedge Q)$:

$$
\begin{aligned}
(R \wedge P) \longrightarrow (R \wedge Q) &\Longleftrightarrow \sim (R \wedge P) \vee (R \wedge Q) \\
&\Longleftrightarrow (\sim R) \vee (\sim P) \vee (R \wedge Q) \\
&\Longleftrightarrow [(\sim R \vee R) \wedge (\sim R \vee Q)] \vee (\sim P) \\
&\Longleftrightarrow [\mathbf{T} \wedge (\sim R \vee Q)] \vee (\sim P) \\
&\Longleftrightarrow (\sim R \vee Q) \vee (\sim P) \\
&\Longleftrightarrow (\sim R) \vee (\sim P \vee Q) \\
&\Longleftrightarrow (\sim R) \vee (P \longrightarrow Q)
\end{aligned}
$$

- From this we observe that when P is true, Q is false and R is false, then the RHS is true and hence the LHS is true. However, the conditional $(P \rightarrow Q)$ is false. Therefore, the LHS cannot be logically equivalent to $(P \rightarrow Q)$.

Demonstration Examples (Pre-class Q4)

Question

Let $\mathbb{Z}$ be a universal set. Write the negation of the following statement in the simplest form:

$$P := \forall x \left(\frac{x}{27} \in \mathbb{N}\right) \vee \exists y((y - 20 > 0) \longrightarrow (y^4 = 42)).$$

- We first write

$$P \Longleftrightarrow \forall x \left(\frac{x}{27} \in \mathbb{N}\right) \vee \exists y(\sim (y - 20 > 0) \vee (y^4 = 42)).$$

- Thus we have

$$\begin{aligned} \sim P &\Longleftrightarrow \sim \left[\forall x \left(\frac{x}{27} \in \mathbb{N}\right) \vee \exists y(\sim (y - 20 > 0) \vee (y^4 = 42))\right] \\ &\Longleftrightarrow \sim \forall x \left(\frac{x}{27} \in \mathbb{N}\right) \wedge \sim \exists y(\sim (y - 20 > 0) \vee (y^4 = 42)) \end{aligned}$$

- Continue:

$$
\begin{aligned}
\sim P &\Longleftrightarrow \sim \forall x \left(\frac{x}{27} \in \mathbb{N}\right) \wedge \sim \exists y(\sim (y - 20 > 0) \vee (y^4 = 42)) \\
&\Longleftrightarrow \exists x \sim \left(\frac{x}{27} \in \mathbb{N}\right) \wedge \forall y \sim (\sim (y - 20 > 0) \vee (y^4 = 42)) \\
&\Longleftrightarrow \exists x \left(\frac{x}{27} \notin \mathbb{N}\right) \wedge \forall y((y - 2 > 0) \wedge \sim (y^4 = 42)) \\
&\Longleftrightarrow \exists x \left(\frac{x}{27} \notin \mathbb{N}\right) \wedge \forall y((y - 2 > 0) \wedge (y^4 \neq 42))
\end{aligned}
$$

Demonstration Examples (Pre-class Q5)

Question

(a) Show that for all $n \in \mathbb{N} \cup \{0\}$

$$49^n \equiv \begin{cases} 1 \pmod{100} & \text{if } n \text{ is even} \\ 49 \pmod{100} & \text{if } n \text{ is odd} \end{cases}$$

(b) Find the last two digits of 7^{2025}.

(a) We separate into cases of odd and even n.

(i) If n is even, then $n = 2k$ for some integer $k \geq 0$.

$$49^n = 49^{2k} = \left(49^2\right)^k = 2401^k \equiv 1^k \equiv 1 \pmod{100}$$

Therefore, $49^n \equiv 1 \pmod{100}$ for even n.

(ii) If n is odd, then $n = 2k+1$ for some integer $k \geq 0$. Similarly, we have

$$49^n = 49^{2k+1} = 49\left(49^2\right)^k = 49 \cdot 2401^k \equiv 49 \cdot 1^k \equiv 49 \pmod{100}$$

Therefore, $49^n = 49 \pmod{100}$ for odd n.

Demonstration Examples (Pre-class Q5)

Question

(a) Show that for all $n \in \mathbb{N} \cup \{0\}$

$$49^n \equiv \begin{cases} 1 \pmod{100} & \text{if } n \text{ is even} \\ 49 \pmod{100} & \text{if } n \text{ is odd} \end{cases}$$

(b) Find the last two digits of 7^{2025}.

(b) We rewrite $7^{2025} = 7^{2 \cdot 1012 + 1} = \left(7^2\right)^{1012} \cdot 7 = 49^{1012} \cdot 7$. Using the multiplicative property of modulo:

$$\begin{aligned} 49^{1012} \cdot 7 &\equiv 1 \cdot 7 \pmod{100} && \text{using (a)} \\ &\equiv 7 \pmod{100} \end{aligned}$$

Therefore, 7^{2025} leaves a remainder of 7 when divided by 100. This implies that its last two digits must be '07'.

Demonstration Examples (Pre-class Q6)

Question

(a) Prove that for any $n \in \mathbb{Z}$, $n^2 \equiv 1 \pmod 4$ or $n^2 \equiv 0 \pmod 4$.

(b) Show that all right $\triangle ABC$ with integer side lengths a, b, c have integer area.

(a) We simply go through all possible $n \pmod 4$:

- $n \equiv 0 \pmod 4 \implies n^2 \equiv 0^2 \equiv 0 \pmod 4$
- $n \equiv 1 \pmod 4 \implies n^2 \equiv 1^2 \equiv 1 \pmod 4$
- $n \equiv 2 \pmod 4 \implies n^2 \equiv 2^2 \equiv 0 \pmod 4$
- $n \equiv 3 \pmod 4 \implies n^2 \equiv 3^2 \equiv 1 \pmod 4$

Therefore, $n^2 \equiv 1 \pmod 4$ or $n^2 \equiv 0 \pmod 4$ for any $n \in \mathbb{Z}$.

Demonstration Examples (Pre-class Q6)

Question

(a) Prove that for any $n \in \mathbb{Z}$, $n^2 \equiv 1 \pmod 4$ or $n^2 \equiv 0 \pmod 4$.

(b) Show that all right $\triangle ABC$ with integer side lengths a, b, c have integer area.

- What is the connection with (a)?
  - ▷ squaring numbers & triangle – Pythagorean theorem!

(b) Suppose c is the hypotenuse, giving us the well-known result:

$$a^2 + b^2 = c^2 \implies a^2 + b^2 \equiv c^2 \pmod 4$$

$$\implies \begin{cases} 1 + 0 \equiv 1 \pmod 4 \\ 0 + 1 \equiv 1 \pmod 4 \\ 0 + 0 \equiv 0 \pmod 4 \end{cases}$$

Here we consider all possible cases due to (a), and we discard the case $a^2, b^2 \equiv 1 \pmod 4$.

Therefore, one of a, b must be congruent to 0 or 2 (modulo 4), and thus one of a, b is even. This gives an integer area $= ab/2$.